

9 (a) <u>thin / fine</u> metal wire lay-out shown as a grid encased in plastic	B1 B1 B1 [3]
(b) (i) gain (of amplifier)	B1 [1]
(ii) for $V_{OUT} = 0$, then $V^+ = V^-$ or $V_1 = V_2$ $V_1 = (1000/1125) \times 4.5$ $V_1 = 4.0\text{ V}$	C1 C1 A1 [3]
(iii) $V_2 = (1000 / 1128) \times 4.5$ $= 3.99\text{ V}$ $V_{OUT} = 12 \times (3.99 - 4.00)$ $= (-) 0.12\text{ V}$	C1 A1 [2]
10 strong / large (uniform) magnetic field nuclei precess / rotate about field direction (1) radio frequency pulse at Larmor frequency (1) causes resonance / nuclei absorb energy on relaxation / de-excitation, nuclei emit r.f. pulse pulse detected and processed (1) non-uniform field superposed on uniform field allows position of resonating nuclei to be determined allows for location of detection to be changed (1) (six points, 1 each plus any two extra – max 8)	B1 B1 B1 B1 B1 B1 B1 [8]